

OPENLEAVE · HOW IT WORKS

From statute to answer.

The four artifacts of Rules as Code — the raw rule, the coded rule, the user's facts, and the system's result — traced through one real determination from the working prototype.

JULY 2026 · every excerpt is real; every result was produced by running the engine

§ 1

RAW RULES

*Thresholds, date arithmetic, a
piecewise formula — closed-texture
law, the kind that encodes faithfully.*

What the law *says*.

The running example: a Minnesotan hired March 2025, full time at \$1,100/week, 85-person employer, bonding leave starting September 1, 2026. Two rules govern it.

"The term 'eligible employee' means an employee who has been employed — (i) for at least 12 months by the employer...; and (ii) for at least 1,250 hours of service with such employer during the previous 12-month period... [excluding] any employee... at a worksite at which such employer employs less than 50 employees [within] 75 miles..."

29 U.S.C. § 2611(2) — Family and Medical Leave Act (excerpt)

"The weekly benefit amount is 90% of wages that do not exceed 50% of the state's average weekly wage, plus 66% of wages between 50% and 100% of that wage... capped at the state average weekly wage. Nearly all employers are covered, regardless of size."

Minn. Stat. ch. 268B — Minnesota Paid Leave (summarized excerpt)

§ 2

CODED RULES

Each statutory condition becomes a Finding that carries its citation — compare clause-by-clause with § 1.

`openleave/regimes/fmla.py`

The same law, *executable*.

```
result.findings = [  
    Finding(  
        key="tenure",  
        description="Employed by this employer for at least 12 months",  
        met=facts.tenure_months >= 12,  
        citation=Citation("29 U.S.C. § 2611(2)(A)(i)"),  
    ),  
    Finding(  
        key="hours",  
        description="At least 1,250 hours of service in the previous 12 months",  
        met=facts.employee.hours_last_12mo >= min_hours,  
        citation=Citation("29 U.S.C. § 2611(2)(A)(ii)"),  
    ),  
    Finding(  
        key="worksite",  
        description="Employer has 50+ employees within 75 miles of the worksite",  
        met=facts.worksite_headcount >= min_headcount,  
        citation=Citation("29 U.S.C. § 2611(2)(B)(ii)"),  
    ),  
]
```

§ 2

CONTINUED

Rates and caps are never hard-coded: effective-dated parameters enable “the law as of any date” — and are what the AI amendment pipeline maintains.

Formulas transcribe; *numbers stay data.*

THE § 268B.06 BENEFIT FORMULA — A DIRECT TRANSCRIPTION OF THE STATUTE'S PIECEWISE STRUCTURE

```
def _weekly_benefit(aww, saww):          # Minn. Stat. § 268B.06
    half = 0.5 * saww
    benefit = 0.9 * min(aww, half)        # 90% up to 50% of SAWW
    if aww > half:
        benefit += 0.66 * (min(aww, saww) - half)  # 66% between 50–100%
    if aww > saww:
        benefit += 0.55 * (aww - saww)           # 55% above
    return round(min(benefit, saww), 2)      # capped at SAWW
```

EFFECTIVE-DATED STATUTORY PARAMETERS — OPENLEAVE/PARAMETERS.JSON

```
"mn.saww":          [ ["2026-01-01", 1423.00] ]
"fmla.min_hours":   [ ["1993-08-05", 1250] ]
"ny.saww":          [ ["2025-01-01", 1757.19], ["2026-01-01", 1839.34] ]
```

§ 3

FACT INPUT

*The system — not the user —
figures out which laws apply.*

What the user *tells the system*.

The caller — an HR platform, a payroll system, a chatbot, or a person using the web checker — submits only facts. No legal knowledge required to ask.

```
POST /determinations
{
  "facts": {
    "employee": {
      "work_state":      "MN",          ← where they work
      "hire_date":      "2025-03-01",  ← when they were hired
      "hours_last_12mo": 1400,          ← how much they work
      "average_weekly_wage": 1100      ← what they earn
    },
    "employer": { "total_employees": 85 }, ← how big the employer is
    "event":      { "type": "bonding", "start": "2026-09-01" } ← what's happening
  }
}
```

§ 4

RESULT

*\$896.76 = 90% of the first
\$711.50 (half the MN SAWW)
+ 66% of the remaining \$388.50.
The arithmetic is reconstructible
from the response.*

The determination, *reasoning attached.*

FMLA (federal)	eligible: true
✓ 12 months' employment – 18.0 months	[29 U.S.C. § 2611(2)(A)(i)]
✓ 1,250 hours of service – 1,400 hours	[29 U.S.C. § 2611(2)(A)(ii)]
✓ 50+ employees / 75 mi – 85 employees	[29 U.S.C. § 2611(2)(B)(ii)]
→ 12 weeks · job-protected · unpaid	[29 U.S.C. § 2614]
Minnesota Paid Leave	eligible: true
✓ wage credits ≥ ~\$3,922 – \$57,200	[Minn. Stat. § 268B.04]
✓ covered employment (no size minimum)	[Minn. Stat. § 268B.01]
→ 12 weeks · job-protected · \$896.76/week	[Minn. Stat. § 268B.06]
interactions	
· leaves run concurrently when properly designated	
· employer may not force substitution of employer paid leave	[DOL guidance 2026]

Condensed from the engine's actual JSON response — every conclusion points back to a § 1 citation, closing the loop.

§ 5

TWO VARIATIONS

Open-textured law is flagged and routed to a person — never resolved by the machine.

Honest about *no*, and about *not knowing*.

SAME FACTS, ONLY 900 HOURS WORKED

FMLA

eligible: false

✓ 12 months' employment

x 1,250 hours of service

hours in previous 12 mo: 900

[29 U.S.C. § 2611(2)(A)(ii)]

✓ 50+ employees / 75 mi

When the answer is no, the tree shows exactly which condition failed — and which still passed.

LEAVE REASON: OWN HEALTH CONDITION

FMLA

eligible: null

✓ tenure · hours · worksite

⚠ "serious health condition"

requires medical certification;

not determinable by rule

[29 C.F.R. § 825.113]

Not yes, not no: the open-textured question is routed to a human, with the governing regulation attached.

§ 6

THE CHAIN

Text to code to answer — with
authority attached at every step.

Every stage *cites its neighbor*.



THE MAINTENANCE LOOP *When the raw rule changes, the amendment pipeline proposes the § 2 diff — LLM-drafted, regression-gated against pinned historic determinations, attorney-approved — so the chain stays closed as the law moves.*

Prototype illustration. Statutory excerpts abridged; parameter values are approximations pending counsel review. Decision support, not legal advice.

IN WITNESS WHEREOF,

The law, the code, the facts,
the answer — one chain.

OPENLEAVE · leave law, as code

github.com/geneweng/cc_rules_as_code · see `how-it-works.md` for the full walkthrough